Airline Operational Reliability

Investigation priorities for Southwest Airlines (WN)

Decision requested: prioritize a DEN reliability review, with PHX as the largest route workload and SFO as a high-rate route; run a separate DAL disruption review. Treat the dashboard as a prioritization tool, then validate causes with operational records before approving staffing, schedule or capacity changes.

WN annual indicator	Observed value
Flight records	1,391,885
Arrival delay rate: 15+ minutes	21.40%
Cancellation / diversion rate	0.85% / 0.23%
Mean delay among arrivals delayed 15+ minutes	56.98 minutes
15+ minute delay burden per valid arrival	12.19 minutes

Three priorities

1. DEN reliability. 90,439 flight records; 30.61% of valid arrivals delayed at least 15 minutes, 9.22 percentage points above the WN benchmark using unrounded values. Its 1,580,610 delay minutes warrant investigation; they are not proven avoidable minutes.

2. Route-specific diagnosis. DEN-PHX has the largest observed DEN route delay total (65,457 minutes). DEN-SFO has a 41.77% delay rate and 24.60 delay minutes per valid arrival, versus DEN averages of 30.61% and 17.64 minutes.

3. DAL disruption preparedness. DAL recorded 1,017 cancellations (1.46% of 69,423 records), the largest WN origin cancellation count in the reviewed station table. Weather accounts for 724 reported cancellations; November also has a substantial NAS component.

Scope and decision limits

The database contains 7,001,619 US domestic reporting-carrier flight records for 2025; WN is the focus subset. This is a portfolio analysis of public BTS data. No passenger, cost, revenue, crew or ground-handling records were available. No financial savings or causal effect is estimated.

DEN: workload and route risk

DEN has 89,623 valid arrival records and 27,435 arrivals delayed at least 15 minutes. Mean delay in that delayed subset is 57.61 minutes; the delay burden is 17.64 minutes per valid arrival. Late Aircraft and Air Carrier account for **84.30% of total delayed-arrival minutes**.

Reported category	Minutes	Share of arrival delay
Late Aircraft	856,224	54.17%
Air Carrier	476,175	30.13%
National Aviation System	221,388	14.01%
Weather	23,565	1.49%
Security	2,338	0.15%
Not allocated to reported causes*	920	0.06%

*Reconciliation gap, not an additional proven cause. Rounded shares may not sum to exactly 100%. Reported cause coverage is 99.9418%.

Annual DEN route	PHX	SFO
Valid arrival records	3,878	1,537
Delayed arrivals: 15+ minutes	1,149	642
Delay rate	29.63%	41.77%
Mean delay in delayed subset	56.97 min	58.90 min
Total delay in delayed subset	65,457 min	37,817 min
Burden per valid arrival	16.88 min	24.60 min
Late Aircraft / Air Carrier / NAS share	52.71% / 31.71% / 13.41%	43.84% / 27.96% / 24.76%

Interpretation. PHX is a workload priority, rather than an unusually high delay-rate route relative to DEN. SFO is 11.16 percentage points above DEN and has approximately 39.46% greater burden per valid arrival. Its higher NAS share supports investigating traffic-management and network restrictions; it does not locate the restriction at SFO.

Concentration. The ten highest-load DEN routes contribute 433,099 minutes (27.40% of DEN total); PHX, LAS and DAL together contribute 10.85%. A review limited to those three routes would cover only a minority of the total burden. Compare month and departure block before proposing a network-wide change. These are descriptive rankings, not significance tests.

DAL: event concentration matters

DAL recorded 1,017 cancellations in 2025. The reported categories were Weather 724 (71.19%), Air Carrier 193 (18.98%), NAS 100 (9.83%) and Security 0. These labels do not establish that weather occurred locally at DAL.

Month	Flights	Cancelled	Rate	Weather	NAS
Jan	5,594	289	5.17%	272	2
Feb	5,197	22	0.42%	6	5
Mar	6,079	78	1.28%	46	3
Apr	5,907	126	2.13%	107	4
May	6,011	112	1.86%	84	2
Jun	6,039	50	0.83%	23	3
Jul	6,179	83	1.34%	58	3
Aug	5,561	27	0.49%	10	1
Sep	5,356	7	0.13%	2	2
Oct	5,930	49	0.83%	34	3
Nov	5,644	153	2.71%	72	70
Dec	5,926	21	0.35%	10	2

January 9: a validated event case

Of 177 reported flights, 135 were cancelled (76.27%): Weather 121 and Air Carrier 14. That day represents **46.71% of January cancellations** and 13.27% of annual DAL cancellations. January as a whole represents 28.42% of the annual total. The event supports a focused preparedness review; it does not explain every January cancellation.

November: a different reported mix

November has 153 cancellations: Weather 72, NAS 70 and Air Carrier 11. NAS therefore accounts for 45.75% of November cancellations. Do not extend January's weather-led explanation to November without reviewing the affected dates, destinations and restrictions.

Required evidence: weather observations and forecasts available at decision time; airport and traffic-management restrictions; de-icing and ground-handling logs; aircraft and crew plans; cancellation decision timestamps. Use these to locate constraints and distinguish the event from recovery execution.

From findings to measurable action

Priority / proposed owner	Next action and evidence	Success measure
DEN rotation review Network Operations + DEN station	Trace prior legs and planned/actual turnaround for PHX and other high-load routes. Join aircraft, crew, maintenance and handling records.	Delay rate, conditional severity and burden per valid arrival; compare matched route/month/time groups.
DEN-SFO network review Network Operations + ATC liaison	Identify dates and blocks behind NAS minutes. Check restrictions at origin, destination and en route; account for flight mix.	NAS minutes per valid arrival and delay rate in comparable operating conditions.
DAL disruption review Station Operations + Network Control	Reconstruct Jan 9 and November events. Check forecast lead time, cancellation timing and recovery constraints before selecting interventions.	Cancellation rate by comparable event severity; recovery time after restrictions lift, once a baseline and timestamps are defined.
Reporting control BI / Data owner	Investigate the 14 cause-total exceptions and one missing regular arrival. Retain source values; reconcile each new monthly load.	Coverage and exception counts by selected period; source/target row-count agreement.

Implementation sequence

First: nominate owners and obtain operational evidence. **Then:** compare matched flights and document the mechanism. **Finally:** pilot a targeted intervention and track both delay and cancellation outcomes. These are proposed next steps, not implemented operational changes. No arbitrary improvement target or resource increase is assumed.

Measurement and data assurance

Delay rate = arrivals delayed 15+ minutes / records with known arrival delay; early arrivals remain in the denominator. Cancellation and diversion rates use all flight records. The 15+ minute burden metric sums only that delayed subset and divides by valid arrivals; it is not mean signed delay. Percentages are calculated from counts, not averaged across months. Carrier comparisons retain the selected period.

One YX flight has missing arrival delay despite neither cancellation nor diversion; it is preserved as NULL. Fourteen WN flights have 3,569 arrival-delay minutes but five zero cause values (75 in September, 297 in October, 3,197 in November). WN coverage is 99.9787%; DEN coverage is 99.9418%. Small annual gaps can be material in narrow selections.

Source and reproducibility

BTS Reporting Carrier On-Time Performance, 2025 monthly extracts: https://transtats.bts.gov/PREZIP/ . Calculations derive from the author's reconciled SQL Server snapshot and Power BI outputs reviewed in October 2026. Public source revisions may change later downloads. Technical archive documents schema, ingestion, DAX, definitions and reported local tests. Clean-database schema and July ingestion tests passed; the packaged files were not rerun on a separate server.